Feature selection techniques

1) Filter methods:

→ IG

→ Chi-square Test

→ Correlation coef.

2) Wrapper methods:

→ Recursive feature elimination.

→ Genetic algorithms

3) Embedded methods:

→ Decision Tree

A	B	C	D	E	T

CHI-SQUARE TEST

STUDY BUDDY

$Z = \frac{x-\mu}{\sigma}$

The Square of Normal Variate is known as Chi-Square Variate.

CHI-SQUARE TEST

STUDY BUDDY

$z = \left(\frac{x-\mu}{\sigma}\right)^2$ → The Square of Normal Variate is known as Chi-Square Variate.

↳ Large - 50

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Coin - 20

H	T
10	10
7	13

Dice - 100

1	2	3	4	5	6
10	15	17			

$\chi^2 < \chi_t$

Table - Dy. - 5%

$(7-1)(3-1)$

CHI-SQUARE TEST

STUDY BUDDY

Q. Table shows No. of Aircraft Accident occured during days of week. Test wether accidents are uniformly distributed.

Days	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Total
No. of Accident	14	18	12	11	15	14	14	98

[df - 6, $\chi^2_{0.05} = 12.592$]

Q. Table shows No. of [illegible] during days of week. Test whether accidents are uniformly distributed. — Buddy

Days	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Total
No. of Accident	14	18	12	11	15	14	14	98

$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$ Let Accidents be unif. dist. [df - 6, $\chi^2_{0.05} = 12.592$]

$\frac{98}{7} = 14$

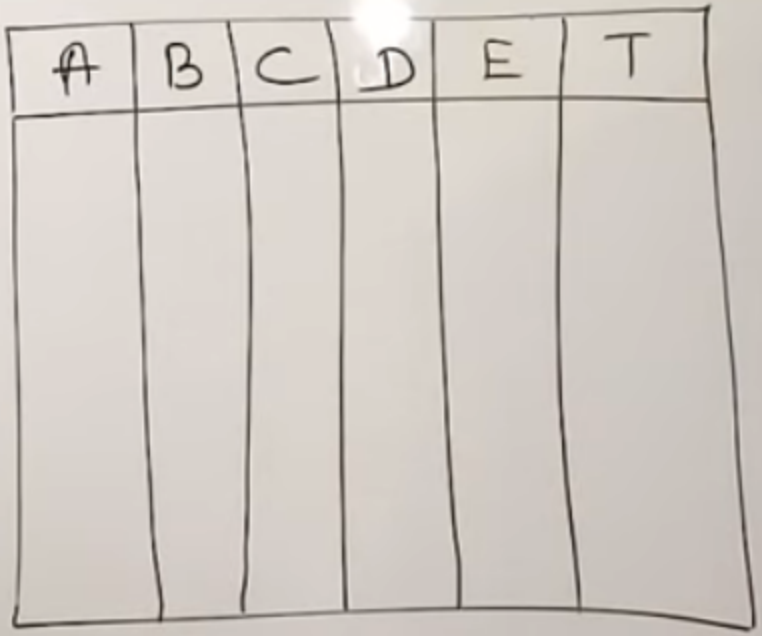
O	E	O-E	$(O-E)^2$	$\frac{(O-E)^2}{E}$
14	14	0	0	0
18	14	4	16	1.14
12	14	-2	4	0.29
11	14	-3	9	0.64
15	14	1	1	0.07
14	14	0	0	0
14	14	0	0	0

$\chi^2 < \chi^2_{0.05}$

The Hyp. is Accept..

(2:14)

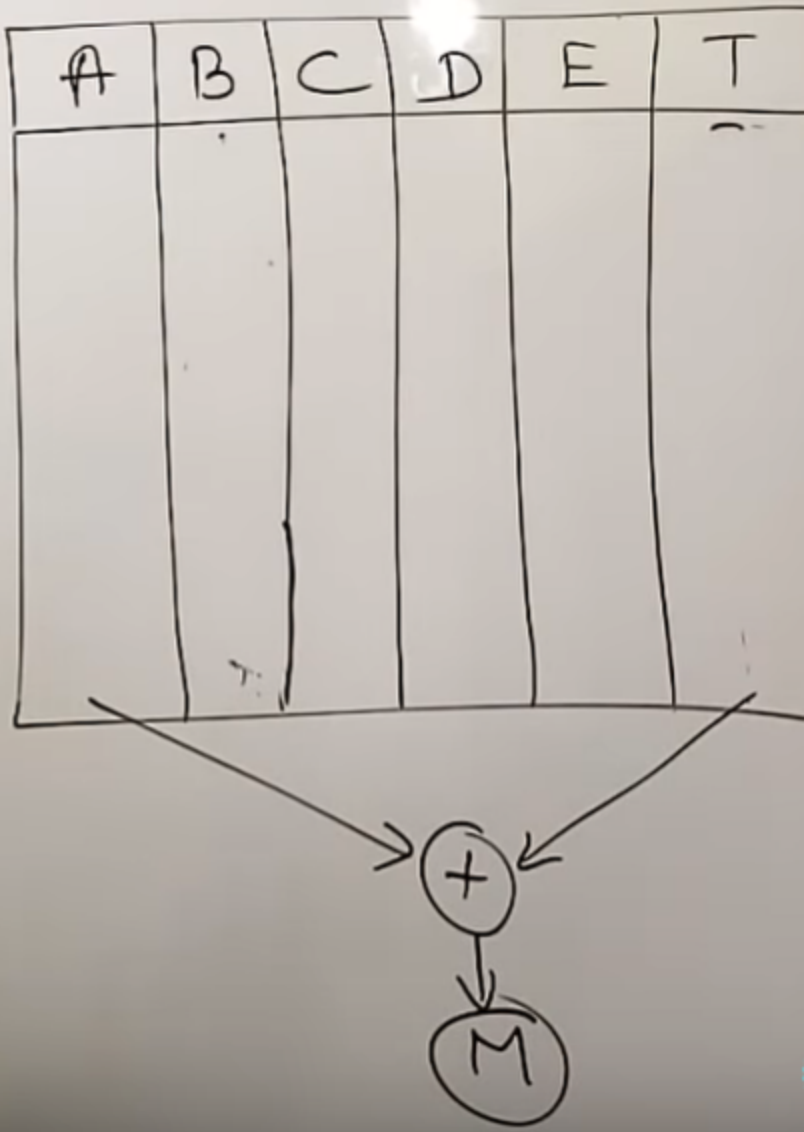
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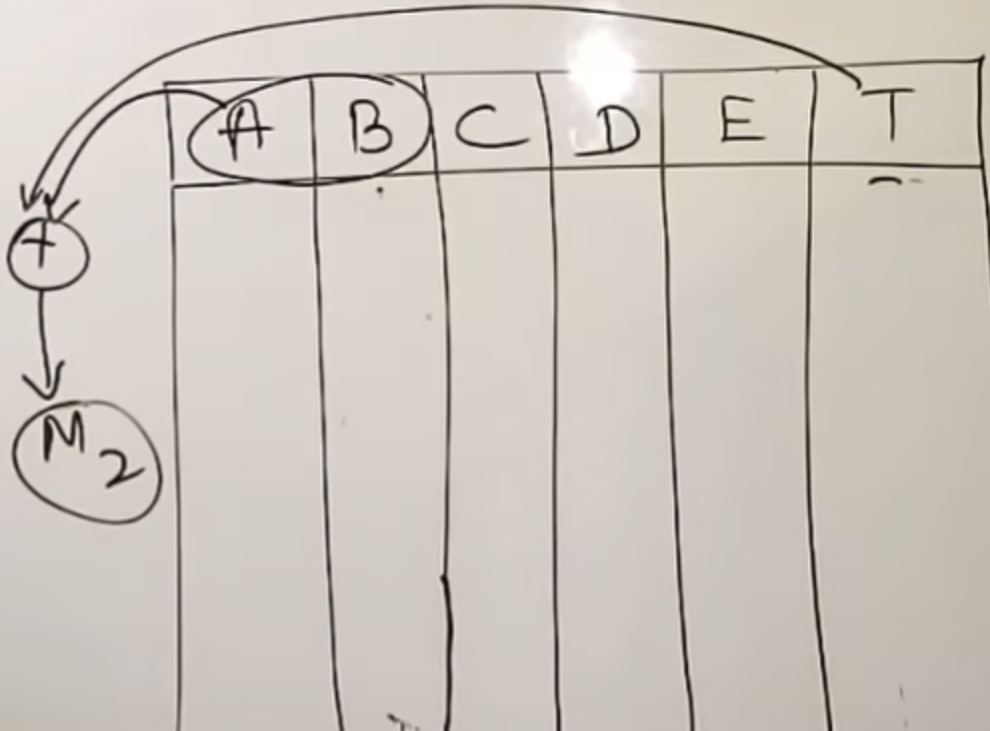


A	B	C	D	E	T
	1				P
	2				F
	3				F
	4				P
	5				F
	6				P
	7				F
					P

tion.

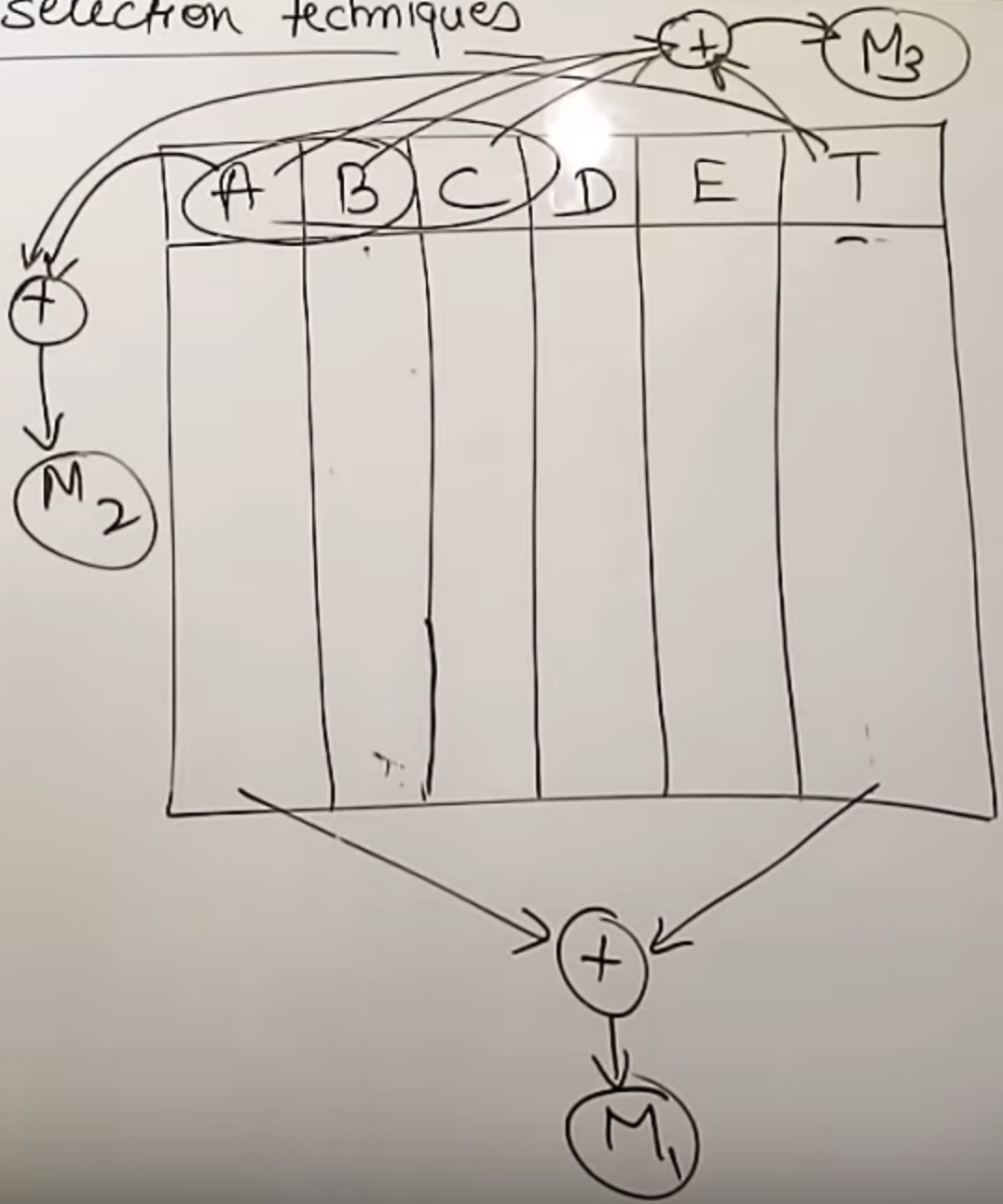
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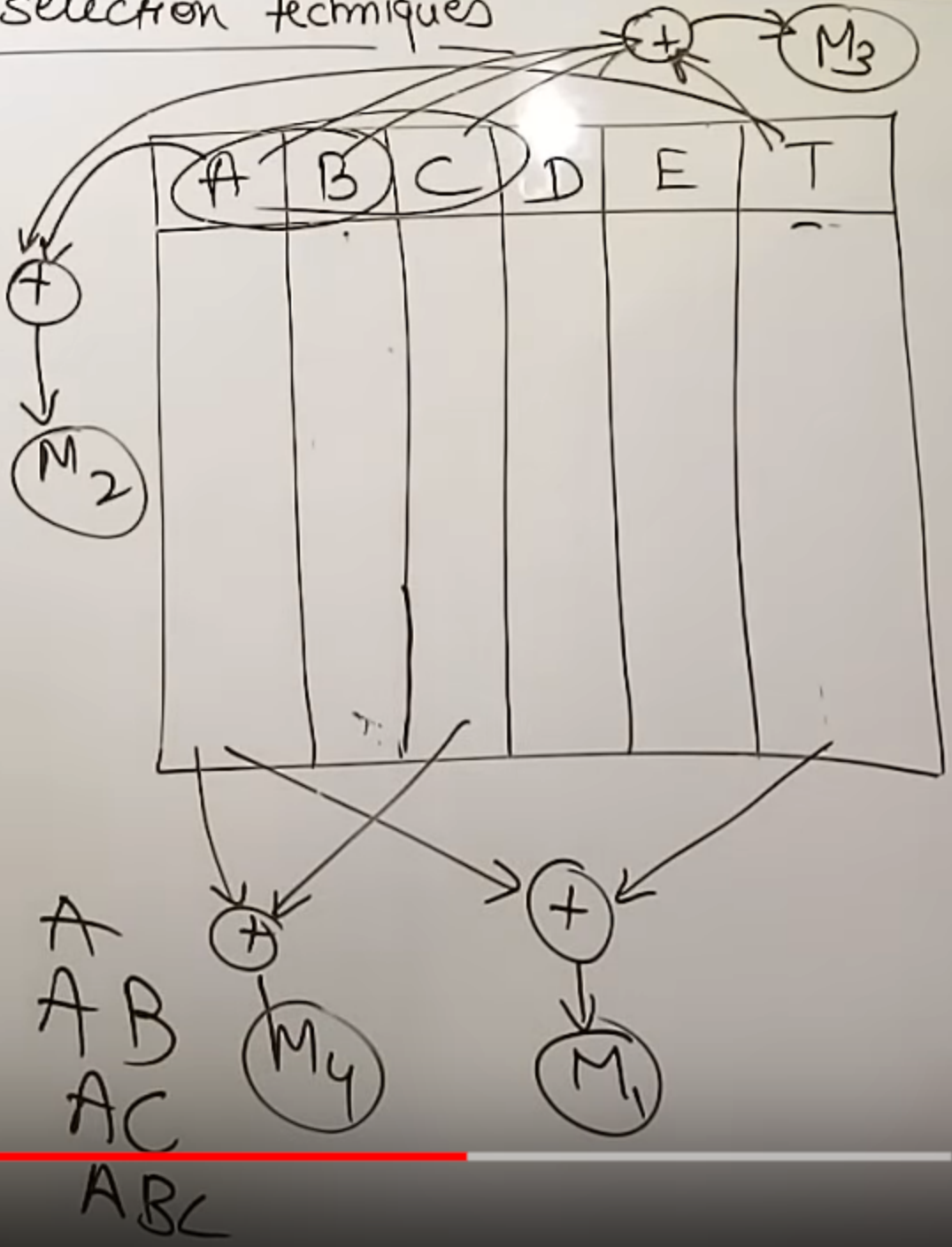
eature selection techniques



limination.

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